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Studierichting: Informatica
Oefeningenreeks 5A nr. 7.6

$$\int_0^{\frac{\pi}{6}} \frac{dx}{\cos x}$$

$$t = \sin x \left(dt = \cos x dx,\ dx = \frac{dt}{\cos x}\right)$$

$$\begin{cases} x = 0 \to t = \sin 0 = 0 \\ x = \frac{\pi}{6} \to t = \sin\left(\frac{\pi}{6}\right) = \frac{1}{2} \end{cases}$$

$$= \int_0^{\frac{1}{2}} \frac{dt}{1-t^2} = \frac{A}{1-t} + \frac{B}{1+t}$$

$$A = \frac{1}{1+t}\Big|_{t=1} = \frac{1}{2}$$

$$B = \frac{1}{1-t}\Big|_{t=-1} = \frac{1}{2}$$

$$\Rightarrow \int_0^{\frac{1}{2}} \frac{\frac{1}{2}}{1-t} dt + \int_0^{\frac{1}{2}} \frac{\frac{1}{2}}{1+t} dt$$

$$= -\frac{1}{2} \ln|1-t| + \frac{1}{2} \ln|1+t| + C$$

$$= \frac{1}{2}\left(-\ln|1-t| + \ln|1+t|\right) + C = \frac{1}{2} \ln\left|\frac{1+t}{1-t}\right|$$

$$= \left[\frac{1}{2} \ln\left(\frac{1+t}{1-t}\right)\right]_0^{\frac{1}{2}}$$

$$= \frac{1}{2} \ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right) - \frac{1}{2} \ln\left(\frac{1+0}{1-0}\right)$$

$$= \frac{1}{2} \ln\left(\frac{\frac{3}{2}}{\frac{1}{2}}\right) - 0$$

$$= \frac{1}{2} \ln 3$$

References